



## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 97.89%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 98.93%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.43%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 2.11%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 97.89%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4682           | FN 101            |
| Actual: Normal | FP 0              | TN 13066          |

### 2b. Random Forest: Attack Classification

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 100.00% | Share of actual attacks successfully flagged         |
| F1-Score                  | 100.00% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 100.00% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 0.00%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 100.00% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 4682            | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Hybrid Cascaded

| Metric                    | Value   | Description                                          |
|---------------------------|---------|------------------------------------------------------|
| Precision                 | 100.00% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 97.89%  | Share of actual attacks successfully flagged         |
| F1-Score                  | 98.93%  | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.43%  | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.00%   | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 2.11%   | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 97.89%  | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 100.00% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 4682           | FN 101            |
| Actual: Normal | FP 0              | TN 13066          |

2d. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 756.4 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 7.5 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2e. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.05% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 3.09% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 3.03% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.24 | 5        | 4            | 2026-09-26 19:35:28 | 2026-09-26 19:38:09 |
| 10.0.0.25 | 5        | 3            | 2026-09-26 19:35:34 | 2026-09-26 19:38:12 |
| 10.0.0.23 | 4        | 3            | 2026-09-26 19:35:27 | 2026-09-26 19:38:08 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action         |
|---------------------|-----------|-------|------------|------------|----------|----------------|
| 2026-09-26 19:35:27 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 19:35:28 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 19:35:34 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined    |
| 2026-09-26 19:35:38 | 10.0.0.23 | DDoS  | ICMP Flood | 90.7%      | Critical | Time Ban (30s) |
| 2026-09-26 19:35:38 | 10.0.0.24 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban (30s) |
| 2026-09-26 19:35:38 | 10.0.0.24 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban       |
| 2026-09-26 19:35:40 | 10.0.0.23 | DDoS  | ICMP Flood | 90.8%      | Critical | Time Ban       |
| 2026-09-26 19:35:45 | 10.0.0.25 | DDoS  | ICMP Flood | 91.3%      | High     | Time Ban (30s) |
| 2026-09-26 19:35:45 | 10.0.0.25 | DDoS  | ICMP Flood | 91.3%      | Critical | Time Ban       |
| 2026-09-26 19:36:08 | 10.0.0.24 | DDoS  | ICMP Flood | 94.6%      | High     | Released       |
| 2026-09-26 19:36:15 | 10.0.0.25 | DDoS  | ICMP Flood | 94.7%      | High     | Released       |
| 2026-09-26 19:36:18 | 10.0.0.24 | DDoS  | ICMP Flood | 94.6%      | Critical | Time Ban (1m)  |
| 2026-09-26 19:36:26 | 10.0.0.25 | DDoS  | ICMP Flood | 94.7%      | Critical | Time Ban (1m)  |
| 2026-09-26 19:36:39 | 10.0.0.23 | DDoS  | ICMP Flood | 92.1%      | Critical | Time Ban (1m)  |
| 2026-09-26 19:37:19 | 10.0.0.24 | DDoS  | ICMP Flood | 94.6%      | Critical | Time Ban (2m)  |
| 2026-09-26 19:38:08 | 10.0.0.23 | DDoS  | ICMP Flood | 93.6%      | Critical | Time Ban (2m)  |
| 2026-09-26 19:38:09 | 10.0.0.24 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban (5m)  |
| 2026-09-26 19:38:12 | 10.0.0.25 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban (5m)  |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.25 | ICMP Flood | 0.6454   | 91.2% | Critical | Phase 2 | 1        | 11s      | 2026-09-26 19:38:12 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6461   | 91.2% | Critical | Phase 2 | 1        | 11s      | 2026-09-26 19:38:09 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6458   | 93.6% | Critical | Phase 2 | 1        | 10s      | 2026-09-26 19:38:08 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6256   | 94.7% | High     | Phase 2 | 1        | 1m 11s   | 2026-09-26 19:37:26 | Ban Expired                               |
| 10.0.0.24 | ICMP Flood | 0.6452   | 94.6% | Critical | Phase 2 | 4        | 1m 11s   | 2026-09-26 19:37:19 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6448   | 92.1% | Critical | Phase 2 | 3        | 1m 12s   | 2026-09-26 19:36:39 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6361   | 94.7% | Critical | Phase 2 | 3        | 11s      | 2026-09-26 19:36:26 | Attack persisted - escalating to time ban |

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.24 | ICMP Flood | 0.6358   | 94.6% | Critical | Phase 2 | 3        | 10s      | 2026-09-26 19:36:18 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6215   | 94.7% | High     | Phase 2 | 1        | 41s      | 2026-09-26 19:36:15 | Ban Expired                               |
| 10.0.0.24 | ICMP Flood | 0.6132   | 94.6% | High     | Phase 2 | 1        | 40s      | 2026-09-26 19:36:08 | Ban Expired                               |
| 10.0.0.23 | ICMP Flood | 0.6181   | 91.2% | High     | Phase 2 | 2        | 41s      | 2026-09-26 19:36:08 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6461   | 91.3% | High     | Phase 2 | 1        | 11s      | 2026-09-26 19:35:45 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6479   | 91.2% | Critical | Phase 2 | 1        | 10s      | 2026-09-26 19:35:38 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6458   | 90.7% | Critical | Phase 2 | 1        | 11s      | 2026-09-26 19:35:38 | Attack persisted - escalating to time ban |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE